The Inseparability of Geometry and Modular Flow: A Structural Inseparability Theorem and Its Implications for the Hard Problem

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Abstract

The hard problem of consciousness presupposes that physical structure and experiential structure can be specified independently. We show that this presupposition is not available in a precise algebraic setting.

Working with Type III_1 von Neumann algebras, we identify emergent geometry as the relevant mathematical proxy for physical structure and Tomita-Takesaki modular flow as the relevant proxy for first-person temporal structure. The resulting question is whether these two sectors can be independently deformed within an admissible algebraic class. We show first that, working within the admissible class selected by $(U3)$, from the proof-relevant axiom package

$$(U1) + (C) + (Amb) + (CN_{\exists}),$$

one obtains a standard half-sided modular inclusion and hence, via the Guido-Longo-Wiesbrock construction, a canonical conformal skeleton. This yields a conformal-level Weak Inseparability theorem. We then show that the correct non-gauge local modulus is the double-interval cross-ratio family, which determines a canonical modular scale datum

$$\{K_b\}_{b\in(0,1)}.$$

Under the additional conditions $(LImp^1)$, $(CR\text{-}nd)$, and (SFa), this datum induces a Type III-compatible QFIM scale shadow

$$h_b = \mathbf{Q}_{\omega,b}(\partial_b K_b, \partial_b K_b)\, db^2,$$

At the abstract level, this yields a full geometric datum consisting of the conformal skeleton together with the scale family. In the worked examples, where the cross-ratio parameter naturally parametrizes spatial position, this datum determines the Lorentzian metric. In this setting, the full geometric datum and the extended modular data are bidirectionally inseparable.

We verify the additional Phase 2 conditions in two structurally distinct models, the Rindler/free-field chiral half-line model and the Virasoro net. Both yield the universal profile

$$h_b = C(1-b)^{-2}db^2,$$

with model dependence entering only through the overall constant. The philosophical consequence is not that the hard problem has been solved in the ordinary production sense, but that its standard formulation rests on a presupposition—independent describability—that is not available in the relevant algebraic setting. The remaining explanatory gap is thereby relocated from a gap between independently specified domains to a structural limit on self-description by observers embedded in $(\mathcal{A}, \omega)$.

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1 The Hidden Presupposition of the Hard Problem

The hard problem of consciousness is usually presented as a question about production: how do physical processes give rise to subjective experience? In that form, the problem appears deeper than the familiar "easy problems" because those concern discrimination, report, integration, control, and other functional capacities, whereas the hard problem concerns the existence of a first-person point of view and the felt character of experience itself. This contrast, introduced most influentially by Chalmers [4], has shaped almost every major debate in contemporary philosophy of mind.

Yet the canonical formulation contains a strong presupposition that is rarely isolated and almost never challenged at the level of formal structure. The presupposition is that physical structure and experiential structure can be specified independently. One first gives a complete account of the physical facts, in third-person terms, and then asks why those facts are accompanied by experience, or by this experience rather than another. Without this prior separability, the production question cannot even be posed in its usual form. The hard problem therefore depends not only on a gap between two domains, but on the prior availability of those domains as independently describable.

This presupposition is shared by positions that otherwise disagree sharply. Functionalism and mainstream physicalism accept it and then attempt to dissolve the residual gap by appeal to organization, role, or supervenience. Panpsychism accepts it as well, but seeks to avoid brute emergence by distributing proto-experiential or experiential properties more fundamentally. Illusionism also inherits the same starting point: it grants the independent specification of physical structure and then argues that what seemed left over on the experiential side is a cognitive misdescription. Integrated Information Theory takes a different route, but still treats consciousness as something to be attached to or read off from an already specified structure by means of an additional quantity. In all of these cases, the debate concerns how the bridge should be built. The deeper assumption that there are two independently specifiable shores is left intact.

The present paper takes aim at that assumption. It does not begin by proposing a new psychophysical law, a new emergent mechanism, or a new primitive consciousness quantity. It begins by asking whether the independent-describability presupposition survives translation into a precise mathematical framework. If "physical structure" is represented not by an unconstrained intuitive picture of matter but by an emergent geometric structure, and if "first-person time" is represented by modular flow in the sense of Tomita-Takesaki theory, then the hard problem can be reformulated as a structural question: can these two sectors be independently varied inside a single admissible algebraic setting?

The central claim of this paper is that, under a precise Type III_1 algebraic framework, they cannot. At the conformal level, emergent causal geometry and conformal modular data are already mutually rigid. With three additional Phase 2 inputs, the same becomes true at the full Lorentzian level: complete geometry and extended modular data cannot be independently deformed. This does not amount to the claim that the hard problem has simply been "solved." It amounts to a more exact and, in one sense, more radical thesis: the standard production form of the hard problem depends on a presupposition that is not available in the relevant mathematical setting.

This is why the main theorem of the paper is an inseparability theorem rather than a reduction theorem. The point is not that one side reduces to the other, nor that one side causally generates the other, but that both emerge as two readings of one underlying algebraic organization. In philosophical terms, the paper argues that the usual mind-body contrast is not fun-

damental enough to underwrite the standard hard-problem question. In mathematical terms, it shows that the admissible deformations of geometry and modular structure do not factor into two independent sectors.

The argument unfolds in three stages. First, we translate the relevant philosophical notions into precise mathematical objects: emergent geometry for the object-side physical structure, and modular flow for first-person temporal structure. Second, we prove the conditional Phase 1 result: working with the carrier specified by $(U3)$, from the proof-relevant axiom package

$$(U1) + (C) + (Amb) + (CN_{\exists})$$

one canonically obtains a conformal skeleton, together with a conformal-level inseparability theorem. Third, we explain how the full metric scale is added by the cross-ratio modular datum and its Type III QFIM shadow, yielding a conditional full inseparability theorem under the additional Phase 2 package

$$(\text{LImp}^1) + (\text{CR-nd}) + (\text{SFa}).$$

Two worked examples, one based on the Rindler/free-field chiral half-line model and one on the Virasoro net, show that the Phase 2 conditions are not merely formal placeholders but are realized in structurally distinct settings.

The resulting philosophical payoff is twofold. First, the hard problem is reframed: the primary error lies not in failing to build a bridge, but in assuming that the bridge must connect independently specifiable domains. Second, the residual explanatory gap is relocated rather than denied. Once independent describability fails, the remaining question is no longer how matter produces experience, but why a finite observer embedded in a single algebraic structure cannot completely internalize the relation between structural invariants and lived articulation. That second issue belongs to the limits of self-description, not to a production gap between two ontologically detached realms.

The rest of the paper is organized as follows. Section 2 identifies the mathematical counterparts of physical structure and first-person time, and formulates the independent-describability issue as a question of independent deformability. Section 3 states the algebraic framework and axioms. Section 4 proves the conditional Phase 1 result: from Paper I's shell interface and dynamical package to a canonical conformal skeleton and Weak Inseparability. Section 5 develops the Phase 2 completion via the cross-ratio modular scale datum and the full inseparability theorems. Section 6 identifies abstract sufficient conditions for the chart axiom $(LImp^1)$: for Diff-covariant nets it is fully discharged, and for non-Diff-covariant nets the remaining burden is isolated to transportability and regularity assumptions. Section 7 verifies the remaining Phase 2 conditions in two worked examples. Section 8 then returns to the hard problem directly and argues that the theorem shows its standard formulation rests on a presupposition not available in the stated algebraic setting, while relocating the remaining explanatory gap to the self-referential horizon of finite observers. Section 9 discusses the broader mathematical and physical significance, and Section 10 states the explicit non-claims and open problems.

1.1 Logical Roadmap

The argument has two phases, each with a clear input-output structure.

Phase 1 (Sections 3–4). Input: the axiom package $(U1) + (C) + (Amb) + (CN_{\exists})$, working with the carrier specified by $(U3)$. Output: a canonical conformal skeleton via the Guido-Longo-Wiesbrock reconstruction, and a conformal-level Weak Inseparability theorem (Theorem 4.8). The key new step is that half-sided modular inclusion is *derived* from the modular-decoherence scaling axiom (C), not assumed.

Phase 2 (Section 5). Input: the additional conditions $(LImp^1)+(CR\text{-}nd)+(SFa)$. Output: a full geometric datum (conformal skeleton + scale shadow) and bidirectional Full Inseparability (Theorem 5.7). The key new object is the cross-ratio modular scale datum $\{K_b\}$ and its Araki-Hessian QFIM shadow h_b.

Section 6 provides abstract sufficient conditions for $(LImp^1)$. Section 7 verifies all Phase 2 conditions in two worked examples. Sections 8–10 draw philosophical consequences, discuss broader significance, and state open problems.

2 Mathematical Identification: Geometry as Physics, Modular Flow as First-Person Time

The philosophical argument of this paper rests on a mathematical theorem. For the theorem to bear on the hard problem, the translation between philosophical concepts and mathematical objects must be made explicit and defended. This section fixes the three identifications that underwrite the rest of the paper: emergent geometry as the mathematical content of physical structure, Tomita-Takesaki modular flow as the mathematical content of first-person time, and independent deformability as the mathematical translation of independent describability.

2.1 Emergent Geometry as Physical Structure

In the standard formulation of the hard problem, "physical structure" is left deliberately broad. It may refer to microphysical states, functional organization, or spacetime geometry, depending on the author. For the present purposes, the relevant notion of physical structure is not an arbitrary state description but the emergent spacetime geometry associated with a quantum-algebraic datum.

The reason for this choice is not merely technical convenience. In the algebraic approach to quantum field theory, spacetime is not a fixed background container. It is recovered from the algebraic relations among observables. The programme initiated by Wiesbrock and developed by Guido, Longo, and others has shown that, under appropriate conditions, the inclusion structure of von Neumann algebras determines a conformal geometry [17, 5], and in favorable cases a full Lorentzian metric. In this setting, "physical structure" is not something given prior to the algebra. It is something read off from the algebra.

This is the correct level of description for the hard problem, because the hard problem is not about the relation between consciousness and some particular physical substrate such as neurons or silicon. It is about the relation between the subjective and the objective as structural categories. If the objective-physical side is itself an emergent reading of algebraic data, then the question of whether it can be independently specified becomes a question about the algebraic structure, not about a particular material realization.

We therefore identify:

$$\text{physical structure} \longleftrightarrow \text{Geom}(\mathcal{N}, \omega),$$

where Geom denotes the emergent geometry extracted from a family $\mathcal{N}$ of von Neumann algebras in a state ω. At the Phase 1 level, this is a conformal skeleton. At the full level, it includes the metric scale.

2.2 Modular Flow as First-Person Time

The second identification is less familiar in the philosophy of mind but is equally well motivated from the algebraic side.

Given a von Neumann algebra $\mathcal{A}$ and a faithful normal state ω, the Tomita-Takesaki theorem [14] produces a canonical one-parameter automorphism group σ_t^ω of $\mathcal{A}$, the modular automorphism group. This group has three properties that make it the natural mathematical counterpart of first-person temporal structure.

First, uniqueness. The modular flow is the unique automorphism group satisfying the KMS condition at inverse temperature $\beta = 1$ with respect to ω. No external clock or background time parameter is needed to define it. It is intrinsic to the pair $(\mathcal{A}, \omega)$.

Second, algebraic relativity. Different choices of algebra $\mathcal{A}$ within the same ambient structure yield different modular flows. This is not a defect. It is the mathematical expression of the fact that different observational perspectives give rise to different temporal experiences. The Unruh effect is the most familiar physical instance [15]: an accelerated observer's modular flow differs from that of an inertial observer, and the difference is experienced as a thermal bath.

Third, structural inseparability at the operator level. The Tomita-Takesaki polar decomposition

$$S_\Omega = J_\Omega \Delta_\Omega^{1/2}$$

is unique. The modular operator Δ and the modular conjugation J are not independently choosable components. One cannot have Δ without J, nor J without Δ. Since Δ^{it} generates the modular flow and J implements the exchange between the algebra and its commutant, this already hints at the inseparability that the main theorem will establish at the geometric level: the temporal structure (Δ) and the spatial/duality structure (J) are two factors of a single operator.

We therefore identify:

$$\text{first-person time} \longleftrightarrow \sigma_t^\omega = \Delta_\omega^{it}(\cdot)\Delta_\omega^{-it}.$$

More precisely, the full modular sector includes not only the one-parameter flow but also the modular Hamiltonian $K_\omega = -\ln \Delta_\omega$, the modular conjugation J_ω, and, at the extended level, the local modular scale data $\{K_b\}$ introduced in Phase 2.

2.3 From Independent Describability to Independent Deformability

With the two identifications in place, the philosophical question can be stated mathematically.

The hard problem, in its standard form, presupposes that one can give a complete specification of the physical facts and then ask, as a further question, why those facts are accompanied by experience. Translated into the present framework, this becomes: can one specify $\mathrm{Geom}(\mathcal{N}, \omega)$ without thereby fixing $\mathrm{Mod}(\mathcal{N}, \omega)$, or conversely?

The natural mathematical formalization of "independent specification" is independent deformability. Two structures are independently describable, in the relevant sense, if and only if there exist admissible deformations of the underlying algebraic data that change one while leaving the other invariant. If no such deformation exists, then specifying one structure automatically constrains the other, and the two cannot be treated as independently given.

This is a precise, falsifiable mathematical question. It does not depend on intuitions about zombies, inverted qualia, or conceivability arguments. It depends on the rigidity properties of the map

$$(\mathcal{N}, \omega) \longmapsto \big(\mathrm{Geom}(\mathcal{N}, \omega),\ \mathrm{Mod}(\mathcal{N}, \omega)\big)$$

within the admissible class defined by the axiom package.
The hard problem, so translated, becomes:

> Does there exist an admissible deformation of $(\mathcal{N}, \omega)$ that changes Geom while preserving Mod, or vice versa?

The inseparability theorems of this paper answer: no. At the conformal level (Phase 1), the answer is unconditional within the axiom class. At the full scale-sensitive level (Phase 2), the answer is conditional on three regularity inputs that are verified in two worked examples.

2.4 What This Translation Does and Does Not Claim

Two caveats are essential.

First, the identification of modular flow with first-person time is a structural correspondence, not an identity claim. We do not assert that modular flow "is" consciousness in any reductive sense. We assert that modular flow is the correct mathematical object that plays the structural role assigned to first-person temporal experience in the hard-problem debate. The correspondence is justified by the three properties listed above: intrinsicness, perspectival relativity, and operator-level inseparability from spatial structure.

Second, the identification of emergent geometry with physical structure is restricted to the level of spacetime geometry. It does not claim to capture all of physics. In particular, it does not address the relation between consciousness and specific material substrates such as neural tissue. The claim is that at the level of structural categories relevant to the hard problem, emergent geometry is the right proxy for the objective-physical side. Whether this extends to more fine-grained physical descriptions is an open question that lies beyond the scope of the present paper.

3 Algebraic Framework and Axioms

We now state the minimal algebraic framework in which the failure of independent deformability can be proved. The point of this section is not to give a general introduction to Type III_1 factors or to algebraic quantum field theory. It is to isolate the exact package of assumptions needed for the Phase 1 theorem and to indicate, in each case, why that assumption belongs in the present hard-problem-led setting.

3.1 Ambient Datum and Admissible Object

We work in the standard form of an ambient von Neumann algebra

$$(\mathcal{H}_U, \mathcal{A}_U, \Omega),$$

where Ω is cyclic and separating for $\mathcal{A}_U$. This is the minimal regularity needed for Tomita-Takesaki theory and is satisfied in all standard AQFT models [6]. Inside $\mathcal{A}_U$ we select an accessible von Neumann subalgebra

$$\mathcal{A}_c \subset \mathcal{A}_U,$$

and define the reference state

$$\omega(X) := \langle \Omega, X\Omega \rangle.$$

The pair

$$(\mathcal{A}_c, \omega)$$

is the basic object of the theory. In the intended applications, $\mathcal{A}_c$ is of Type III_1. This is not an arbitrary technical preference. Type III_1 is the natural habitat of local quantum field theoretic algebras [6], and it is precisely the setting in which modular theory is rich enough to support the present route from algebraic accessibility data to emergent geometry.

The ambient algebra $\mathcal{A}_U$ should be thought of as the unreduced field of possible observables, while $\mathcal{A}_c$ is the algebra actually stabilized or selected by the accessibility criterion. The question of the paper is then asked not about the entire ambient structure at once, but about what can be extracted from the selected pair $(\mathcal{A}_c, \omega)$.

3.2 Inherited Accessibility Conditions

The first two carrier assumptions are motivated by the HAFF setting [8], but are stated here as independent Paper I inputs.

(U1) Faithfulness. The restriction of the reference state to the accessible algebra is faithful:

$$\omega|_{\mathcal{A}_c} \text{ is faithful.}$$

Equivalently, Ω is cyclic and separating for $\mathcal{A}_c$.

This is the minimal condition that allows modular theory to be applied to $(\mathcal{A}_c, \omega)$. In the present philosophical translation, $(U1)$ says that the allegedly "first-person" sector is not attached to a degenerate algebraic viewpoint. The accessible algebra must support a genuine intrinsic modular structure.

(U3) Stable carrier. The accessible algebra is supplied as a von Neumann subalgebra that is faithful and forward-stable under the decoherence semigroup:

$$\mathcal{E}_t(\mathcal{A}_c) \subset \mathcal{A}_c, \qquad \forall t \geq 0,$$

Here $\{\mathcal{E}_t\}_{t\geq 0}$ is the decoherence semigroup acting on the ambient algebra: a strongly continuous one-parameter semigroup of normal completely positive unital maps on $\mathcal{A}_U$ satisfying the Markov property $\mathcal{E}_0 = \mathrm{id}$ and $\mathcal{E}_{s+t} = \mathcal{E}_s \circ \mathcal{E}_t$. Its physical role is to model the progressive loss of coherence between the accessible algebra and its environment. No identity $\mathcal{A}_c = \mathrm{Fix}(\mathcal{E})$, and no uniqueness or maximality property, is assumed here. (For the motivating construction, see [8].)

The role of $(U3)$ is only to specify the carrier on which the modular construction is performed. The Phase 1 proofs use forward stability, not a selection or uniqueness theorem. Canonical selection of $\mathcal{A}_c$ from a larger registry of accessible effects, probe channels, records, or exact fixed points is a separate interface problem.

Throughout this paper, we work with a carrier satisfying $(U3)$. No claim below promotes that supplied carrier to the unique accessible algebra.

3.3 The New Dynamical Axiom

The crucial new input is the scaling relation between decoherence and modular evolution.

(C) Modular-decoherence scaling covariance. For all $s \in \mathbb{R}$ and $t \geq 0$,

$$\mathcal{E}_t \circ \mathrm{Ad}\,\Delta_{\mathcal{A}_c}^{is} = \mathrm{Ad}\,\Delta_{\mathcal{A}_c}^{is} \circ \mathcal{E}_{e^{2\pi s}t}.$$

Here $\mathrm{Ad}\,\Delta_{\mathcal{A}_c}^{is}$ denotes the outer conjugation action of the modular unitary on the ambient operator algebra.

This is the only genuinely new dynamical axiom introduced in Phase 1. Its content is that modular evolution does not merely preserve the accessible algebra tautologically. It rescales the effective decoherence depth. In the language of the hard problem, (C) is the first place where the object-side and act-side structures are explicitly tied together. It is therefore the real replacement for the earlier non-scrambling intuition.

Remark 3.1 (Physical plausibility of (C)). Axiom (C) holds whenever the modular flow of the accessible algebra acts as a dilation on the decoherence time parameter. This is the case in the Bisognano-Wichmann setting, where modular flow is a Lorentz boost and decoherence depth scales with the boost parameter, and more generally in any Rindler-type wedge algebra. The axiom is a nontrivial dynamical input: it ties the intrinsic modular time of the accessible algebra to the extrinsic decoherence structure of the ambient algebra. Its domain of validity beyond the HAFF setting [8] is an open question.

Remark 3.2 (Finite-dimensional bounded-generator obstruction). There is no nontrivial finite-dimensional realization of (C) with a bounded generator. If $\mathcal{E}_t = e^{t\mathcal{L}}$, differentiating $\mathcal{E}_t\sigma_s = \sigma_s\mathcal{E}_{e^{2\pi s}t}$ at $t = 0$ gives

$$\sigma_s^{-1}\mathcal{L}\sigma_s = e^{2\pi s}\mathcal{L}.$$

Because the modular action is isometric, similarity by σ_s preserves the operator norm. Hence $\|\mathcal{L}\| = e^{2\pi s}\|\mathcal{L}\|$ for every s, forcing $\mathcal{L} = 0$. Thus a nontrivial model must leave the finite-dimensional bounded-generator regime; this remark is an obstruction, not an existence claim for the Type III case.

3.4 Ambient Shell Realization

To turn (C) into a geometric statement, one needs an external family on which the rescaling can act.

(Amb) Operational shell interface. Fix $\varepsilon > 0$. For O in the norm unit ball $\mathrm{Ball}(\mathcal{A}_U)$, define

$$d_\tau(O) := \sup_{0\le t\le\tau} \|\mathcal{E}_t(O) - O\|, \qquad \mathcal{S}_\varepsilon(\tau) := \{O \in \mathrm{Ball}(\mathcal{A}_U) : d_\tau(O) \le \varepsilon\},$$

and supply the ambient shell family by

$$\mathcal{A}(\tau) := W^*\big(\mathcal{A}_c \cup \mathcal{S}_\varepsilon(\tau)\big), \qquad \tau > 0.$$

The ambient modular action $\sigma_s = \mathrm{Ad}\,\Delta_{\mathcal{A}_c}^{is}$ is assumed to act on $\mathcal{A}_U$ by isometric $*$-automorphisms. The following closure and standardness properties are interface inputs:

1. for every $\tau > 0$,
$$\mathcal{A}_c \subset \mathcal{A}(\tau) \subset \mathcal{A}_U;$$
2. if $\tau_1 < \tau_2$, then
$$\mathcal{A}(\tau_1) \supset \mathcal{A}(\tau_2);$$
this nesting also follows directly from $d_{\tau_1} \le d_{\tau_2}$;
3. the accessible algebra is the core limit:
$$\mathcal{A}_c = \bigcap_{\tau>0} \mathcal{A}(\tau);$$
4. Ω is cyclic and separating for every $\mathcal{A}(\tau)$.

The point of (Amb) is to make shell membership and von Neumann closure explicit. It is a Paper I interface input: Foundation I does not derive this membership rule, its core identity, or its standardness. Crucially, modular covariance is not included in (Amb); shell scaling will be derived from the independent dynamical axiom (C).

3.5 Existential Conormality

The final input required for the Phase 1 conformal reconstruction is a second-direction existence condition.

(CN$_\exists$) Existential conormality. There exists some $\mu_0 > 0$ such that, for

$$\mathcal{C}_{\mu_0} := \mathcal{B}'_{\mu_0} \cap \mathcal{A}_c,$$

the vector Ω is cyclic.

The algebras $\mathcal{B}_\mu$ will be constructed from (C) and (Amb) in Section 4; the present point is only that one cyclicity assumption at one scale is enough to close the second direction and trigger the GLW reconstruction.

Philosophically, $(CN_\exists)$ is the minimal extra input needed to pass from a single half-sided modular direction to a full conformal skeleton. It should therefore be read neither as an arbitrary technical patch nor as a hidden second dynamics. It is an existence-type condition ensuring that the accessible algebra contains enough internal relational richness to support the second lightlike direction.

3.6 What Is Explicitly Not Assumed

One older condition is intentionally removed.

(U2) is deleted. The statement that the modular group preserves the accessible algebra,

$$\sigma_t^{\mathcal{A}_c}(\mathcal{A}_c) = \mathcal{A}_c,$$

is automatic from Tomita-Takesaki theory [14] and therefore carries no genuine selection content.

This deletion matters conceptually. The paper does not rest on a tautological "modular stability" assumption. The real content enters through (C), where modular evolution acts outwardly on the shell family rather than merely inwardly on $\mathcal{A}_c$ itself.

It is also worth noting that the stronger exhaustion condition

$$\text{OE}: \quad W^*\Big(\bigcup_{\tau>0} \mathcal{A}(\tau)\Big) = \mathcal{B}(\mathcal{H}_U)$$

is not part of the minimal Phase 1 package. It is only relevant for stronger scalar-core statements that are not needed for the present conformal-level theorem.

3.7 The Minimal Phase 1 Package

We may now summarize the standing assumptions. Throughout the paper we work within the admissible class selected by $(U3)$. The proof-relevant Phase 1 package is:

$$(U1) + (C) + (Amb) + (CN_\exists).$$

The logic of this package is straightforward.

- $(U1)$ makes modular theory available on the accessible algebra.
- (C) ties modular evolution to decoherence depth.
- (Amb) gives that depth a concrete shell realization inside the ambient algebra.
- $(CN_{\exists})$ provides the minimal extra cyclicity needed to reconstruct the second direction and hence the conformal skeleton.

The remainder of Phase 1 consists in proving that this proof-relevant package is already enough to derive a standard half-sided modular inclusion, a canonical GLW Möbius net, and therefore a conformal-level inseparability theorem. That is the task of the next section.

4 From Axioms to Conformal Skeleton (Phase 1)

4.1 From Axiom C to Standard Half-Sided Modular Inclusion

The first structural point of the paper is that the half-sided modular inclusion needed by the Wiesbrock-GLW machinery is not postulated. It is derived conditionally from Paper I's dynamical axiom (C) and shell interface (Amb). In particular, the present route differs sharply from the earlier formulation in which HSMI appeared as an external ansatz. Here it is the output of the Phase 1 assumptions.

Throughout this section we assume $(U1)$, (C), and the explicit shell interface (Amb), and we work with the carrier specified by $(U3)$. We write

$$\sigma_t := \sigma_t^{\mathcal{A}_c}$$

for the modular automorphism group of $(\mathcal{A}_c, \Omega)$ and

$$J := J_{\mathcal{A}_c}$$

for the associated modular conjugation. By (Amb), we are given a decreasing family of ambient shell algebras

$$\mathcal{A}_c \subset \mathcal{A}(\tau) \subset \mathcal{A}_U, \qquad \tau > 0,$$

with

$$\mathcal{A}_c = \bigcap_{\tau>0} \mathcal{A}(\tau),$$

and whose parameter τ is interpreted as a decoherence window.

4.2 Outer Scaling from Modular-Decoherence Covariance

The real content of axiom (C) is that the modular flow of the accessible algebra does not merely preserve $\mathcal{A}_c$ tautologically. Rather, it rescales the ambient shell parameter. This is the first point at which the older $(U2)$-style formulation must be replaced.

Proposition 4.1 (Outer scaling law)**.** *Conditional on the shell interface* (Amb) *and the independent dynamical covariance axiom* (C)*,*

$$\sigma_s(\mathcal{A}(\tau)) = \mathcal{A}(e^{-2\pi s}\tau), \qquad \forall s \in \mathbb{R},\ \tau > 0.$$

Proof. For $O \in \mathrm{Ball}(\mathcal{A}_U)$, axiom (C) and isometry of σ_s give

$$\begin{aligned} d_{e^{-2\pi s}\tau}(\sigma_s O) &= \sup_{0\le t\le e^{-2\pi s}\tau} \|\mathcal{E}_t(\sigma_s O) - \sigma_s O\| \\ &= \sup_{0\le t\le e^{-2\pi s}\tau} \|\mathcal{E}_{e^{2\pi s}t}(O) - O\| \\ &= d_\tau(O), \end{aligned}$$

where the last line uses $u = e^{2\pi s}t$. Thus $\sigma_s \mathcal{S}_\varepsilon(\tau) = \mathcal{S}_\varepsilon(e^{-2\pi s}\tau)$. Since modular flow preserves $\mathcal{A}_c$, applying σ_s to the generating sets and passing to their von Neumann closures proves the equality. □

This result is the correct algebraic replacement for the old non-scrambling intuition: modular time rescales decoherence depth. It is already an inseparability statement in weak form, because the shell parameter and the modular parameter are no longer independently adjustable.

4.3 Outer Standardness

The next observation is simple but essential: the ambient shells are standard for the same vector Ω. This is where the ambient realization is used as a regularity hypothesis rather than as extra dynamics.

Lemma 4.2 (Outer standardness)**.** *For every $\tau > 0$, the vector Ω is cyclic and separating for $\mathcal{A}(\tau)$.*

Proof. Since $\mathcal{A}_c \subset \mathcal{A}(\tau)$ and Ω is cyclic for $\mathcal{A}_c$ by $(U1)$, it is cyclic for $\mathcal{A}(\tau)$. Since $\mathcal{A}(\tau) \subset \mathcal{A}_U$ and Ω is separating for $\mathcal{A}_U$ by the ambient datum (Section 3.1), separation passes to the subalgebra $\mathcal{A}(\tau)$. □

This standardness will allow us to transport the outer shell family into an internal family inside $\mathcal{A}_c$ by Tomita-Takesaki duality.

4.4 The J-Dual Internal Filtration

The key construction is to turn the outer shell family into a family of subalgebras of the accessible algebra itself. This is the point at which the earlier interiorization axiom becomes unnecessary.

For $\mu > 0$, define

$$\mathcal{B}_\mu := J\,\mathcal{A}(1/\mu)'\,J.$$

The next proposition records the properties of this family needed for the HSMI step.

Proposition 4.3 (J-dual internal filtration)**.** *The family $\{\mathcal{B}_\mu\}_{\mu>0}$ satisfies:*

1. *$\mathcal{B}_\mu \subset \mathcal{A}_c$ for all $\mu > 0$;*
2. *if $\mu_1 > \mu_2$, then $\mathcal{B}_{\mu_1} \subset \mathcal{B}_{\mu_2}$;*
3. *Ω is cyclic and separating for each $\mathcal{B}_\mu$;*
4. *the modular group rescales the family according to*

$$\sigma_s(\mathcal{B}_\mu) = \mathcal{B}_{e^{-2\pi s}\mu};$$

5. *the family generates the accessible algebra:*

$$W^*\Big(\bigcup_{\mu>0}\mathcal{B}_\mu\Big)=\mathcal{A}_c.$$

Proof. The inclusion $\mathcal{A}_c\subset\mathcal{A}(1/\mu)$ implies

$$\mathcal{A}(1/\mu)'\subset\mathcal{A}_c',$$

hence

$$\mathcal{B}_\mu=J\,\mathcal{A}(1/\mu)'\,J\subset J\,\mathcal{A}_c'\,J=\mathcal{A}_c,$$

which proves (1).

For (2), if $\mu_1>\mu_2$, then $1/\mu_1<1/\mu_2$, so the monotonicity of the outer shell family gives

$$\mathcal{A}(1/\mu_1)\supset\mathcal{A}(1/\mu_2),$$

hence

$$\mathcal{A}(1/\mu_1)'\subset\mathcal{A}(1/\mu_2)'$$

and therefore

$$\mathcal{B}_{\mu_1}\subset\mathcal{B}_{\mu_2}.$$

For (3), Lemma 4.2 shows that Ω is cyclic and separating for $\mathcal{A}(1/\mu)$. Therefore Ω is also cyclic and separating for $\mathcal{A}(1/\mu)'$. Since $J\Omega=\Omega$ and J is antiunitary, the same holds for

$$J\,\mathcal{A}(1/\mu)'\,J=\mathcal{B}_\mu.$$

For (4), using the Tomita relation

$$\Delta^{is}J=J\Delta^{-is},$$

we compute

$$\sigma_s(\mathcal{B}_\mu)=\Delta^{is}J\,\mathcal{A}(1/\mu)'\,J\Delta^{-is}=J\Delta^{-is}\mathcal{A}(1/\mu)'\Delta^{is}J=J\,\sigma_{-s}\big(\mathcal{A}(1/\mu)\big)'\,J.$$

Now Proposition 4.1 gives

$$\sigma_{-s}\big(\mathcal{A}(1/\mu)\big)=\mathcal{A}(e^{2\pi s}/\mu),$$

so

$$\sigma_s(\mathcal{B}_\mu)=J\,\mathcal{A}(e^{2\pi s}/\mu)'\,J=\mathcal{B}_{e^{-2\pi s}\mu}.$$

Finally, (5) follows from the core limit in (Amb):

$$\mathcal{A}_c=\bigcap_{\tau>0}\mathcal{A}(\tau)\quad\Longrightarrow\quad\mathcal{A}_c'=W^*\Big(\bigcup_{\tau>0}\mathcal{A}(\tau)'\Big).$$

Applying $J(\cdot)J$ yields

$$\mathcal{A}_c=J\mathcal{A}_c'J=W^*\Big(\bigcup_{\tau>0}J\,\mathcal{A}(\tau)'\,J\Big)=W^*\Big(\bigcup_{\mu>0}\mathcal{B}_\mu\Big).$$

□

This construction is the first genuinely new structural step of the Phase 1 chain. The internal family is not imposed from outside. It is obtained functorially from the shell realization and modular data.

4.5 Half-Sided Modular Inclusion as a Theorem

We may now isolate the fixed inclusion that drives the Wiesbrock machinery.

Theorem 4.4 (HSMI from modular-decoherence covariance)**.** *For any fixed $\mu_0 > 0$, the inclusion*

$$(\mathcal{B}_{\mu_0} \subset \mathcal{A}_c, \Omega)$$

is a half-sided modular inclusion, namely

$$\sigma_t(\mathcal{B}_{\mu_0}) \subset \mathcal{B}_{\mu_0}, \qquad \forall t \leq 0.$$

Proof. By Proposition 4.3(3), Ω is cyclic and separating for both $\mathcal{B}_{\mu_0}$ and $\mathcal{A}_c$. By Proposition 4.3(4),

$$\sigma_t(\mathcal{B}_{\mu_0}) = \mathcal{B}_{e^{-2\pi t}\mu_0}.$$

If $t \leq 0$, then $e^{-2\pi t} \geq 1$, hence $e^{-2\pi t}\mu_0 \geq \mu_0$. By Proposition 4.3(2),

$$\mathcal{B}_{e^{-2\pi t}\mu_0} \subset \mathcal{B}_{\mu_0}.$$

Therefore

$$\sigma_t(\mathcal{B}_{\mu_0}) \subset \mathcal{B}_{\mu_0}, \qquad t \leq 0.$$

□

The important point is conceptual as much as technical: in the present framework, HSMI is not a geometric input imported from AQFT. It is the algebraic consequence of a scaling law that ties modular flow to decoherence depth.

4.6 The Wiesbrock-Borchers Positive Generator

Once the fixed inclusion is available, the standard Wiesbrock-Borchers conclusion follows.

Corollary 4.5 (Positive translation generator for the fixed inclusion)**.** *For each fixed $\mu_0 > 0$, there exists a unique positive self-adjoint operator*

$$P_{\mu_0} \geq 0$$

such that, with

$$U_{\mu_0}(a) := e^{-iaP_{\mu_0}}, \qquad a \geq 0,$$

one has

1. $U_{\mu_0}(a)\Omega = \Omega$ *for all* $a \geq 0$*;*
2. *the Borchers covariance relation*

$$\Delta_{\mathcal{A}_c}^{is} U_{\mu_0}(a) \Delta_{\mathcal{A}_c}^{-is} = U_{\mu_0}(e^{-2\pi s}a)$$

holds for all $s \in \mathbb{R}$ *and* $a \geq 0$.

Proof. This is the standard Wiesbrock theorem for a fixed half-sided modular inclusion [17], together with Borchers covariance [2]. □

At this stage we do not claim a pointwise identification between the entire family $\{\mathcal{B}_\mu\}_{\mu>0}$ and the Wiesbrock orbit generated by P_{μ_0}. Such an identification requires an additional normalization or uniqueness argument and is not needed for the Phase 1 main chain.

This completes the first half of Phase 1. We have shown that the accessible algebra canonically carries the half-sided modular structure needed to enter the Möbius reconstruction theorem.

4.7 GLW Conformal Skeleton and Weak Inseparability

The second half of Phase 1 upgrades the fixed half-sided modular inclusion to a full conformal skeleton. The only additional input is an existential conormality condition. Once this is imposed, the Guido-Longo-Wiesbrock reconstruction theorem produces a canonical strongly additive local Möbius net on S^1, and with it the conformal-level version of inseparability.

Throughout this section we assume, in addition to the hypotheses of Section 4.1, the conormality condition $(CN_\exists)$: there exists $\mu_0 > 0$ such that the vector Ω is cyclic for

$$\mathcal{C}_{\mu_0} := \mathcal{B}'_{\mu_0} \cap \mathcal{A}_c.$$

4.8 The Second Direction from Existential Conormality

The role of $(CN_\exists)$ is very precise. It is not used to construct the half-sided inclusion itself. That has already been achieved in Section 4.1. Rather, it upgrades the fixed inclusion to a standard one in the sense required by the GLW theorem.

For $\mu > 0$, define

$$\mathcal{C}_\mu := \mathcal{B}'_\mu \cap \mathcal{A}_c.$$

Proposition 4.6 (The second direction)**.** *Under $(U1)$, (C), (Amb), and $(CN_\exists)$, the family $\{\mathcal{C}_\mu\}_{\mu>0}$ satisfies:*

1. *$\mathcal{C}_\mu \subset \mathcal{A}_c$ and Ω is separating for $\mathcal{C}_\mu$;*
2. *the modular group rescales the family by*
$$\sigma_t(\mathcal{C}_\mu) = \mathcal{C}_{e^{-2\pi t}\mu};$$
3. *for all $t \geq 0$,*
$$\sigma_t(\mathcal{C}_\mu) \subset \mathcal{C}_\mu;$$
4. *if Ω is cyclic for one $\mathcal{C}_{\mu_0}$, then by modular propagation it is cyclic for all $\mathcal{C}_\mu$.*

In particular, the fixed inclusion

$$(\mathcal{B}_{\mu_0} \subset \mathcal{A}_c, \Omega)$$

is standard.

Proof. The inclusion $\mathcal{C}_\mu \subset \mathcal{A}_c$ is immediate from the definition. Separation of Ω follows from $(U1)$, since $\omega|_{\mathcal{A}_c}$ is faithful and faithfulness passes to subalgebras.

For modular covariance,

$$\sigma_t(\mathcal{C}_\mu) = \sigma_t(\mathcal{B}'_\mu \cap \mathcal{A}_c) = \sigma_t(\mathcal{B}_\mu)' \cap \sigma_t(\mathcal{A}_c) = \mathcal{B}'_{e^{-2\pi t}\mu} \cap \mathcal{A}_c = \mathcal{C}_{e^{-2\pi t}\mu},$$

where we used Proposition 4.3(4) and the tautological invariance of $\mathcal{A}_c$ under its own modular group.

If $t \geq 0$, then $e^{-2\pi t} \leq 1$, so the monotonicity of the $\mathcal{B}_\mu$ family implies the reverse monotonicity of the relative commutants:

$$\mathcal{C}_{e^{-2\pi t}\mu} \subset \mathcal{C}_\mu.$$

Thus

$$\sigma_t(\mathcal{C}_\mu) \subset \mathcal{C}_\mu.$$

Finally, $(CN_\exists)$ gives cyclicity for one $\mathcal{C}_{\mu_0}$, and the modular covariance formula propagates this cyclicity along the entire orbit. Since Section 4.1 already gives cyclicity and separation for $\mathcal{B}_{\mu_0}$ and $\mathcal{A}_c$, the inclusion is standard. □

This is the exact point at which the second direction closes. No stronger global conormality assumption is needed for the fixed inclusion that feeds the GLW theorem.

4.9 The GLW Reconstruction Theorem

Once standardness is available, the conformal skeleton appears by a known AQFT theorem. The novelty in the present paper is not the GLW theorem itself, but the conditional construction of its input from the explicit Paper I shell interface and dynamical package.

Theorem 4.7 (Conformal skeleton via Guido-Longo-Wiesbrock)**.** *Fix $\mu_0 > 0$ as in $(CN_\exists)$. Then the standard half-sided modular inclusion*

$$(\mathcal{B}_{\mu_0} \subset \mathcal{A}_c, \Omega)$$

determines a unique strongly additive local Möbius covariant net

$$I \longmapsto \mathcal{A}_{\mathrm{GLW}}(I)$$

on S^1 such that

$$\mathcal{A}_{\mathrm{GLW}}(0, +\infty) = \mathcal{A}_c, \qquad \mathcal{A}_{\mathrm{GLW}}(1, +\infty) = \mathcal{B}_{\mu_0},$$

and Ω is the vacuum vector for the corresponding positive-energy representation of $\mathrm{PSL}(2, \mathbb{R})$.

Proof. This is precisely the Guido-Longo-Wiesbrock reconstruction theorem for standard half-sided modular inclusions [5]. □

Two clarifications are important here.

First, at this stage no separate relative-position hypothesis is needed. Standard HSMI is enough to reconstruct the Möbius net. The previously suspected relative-position bottleneck concerns later identification questions, not the existence of the conformal skeleton itself.

Second, the net produced here is an abstract but canonical conformal object attached to the fixed inclusion. The remaining question of identifying this GLW net pointwise with the pre-given shell filtrations or later cross-ratio families belongs to the next phase of the program.

4.10 The Conformal Skeleton as the Output of Modular Data

Theorem 4.7 already yields the first inseparability statement of the paper. The data entering the standard inclusion are modular data; the output is a conformal geometry in the minimal sense appropriate to one spatial and one temporal dimension, namely a causal/conformal skeleton.

To make this precise, let

$$\mathrm{Mod}_{\mathrm{conf}}(\mathcal{A}_c, \omega)$$

denote the conformal modular sector determined in Phase 1: the modular group of $(\mathcal{A}_c, \Omega)$, the fixed-inclusion positive generator P_{μ_0}, and the corresponding Möbius representation recovered from the standard HSMI. Let

$$\mathrm{Conf}(\mathcal{A}_c, \omega)$$

denote the conformal skeleton, equivalently the GLW net together with its Möbius vacuum representation.

Theorem 4.8 (Conditional Weak Inseparability)**.** *Conditional on the supplied shell interface (Amb), the independent dynamical axiom (C), $(U1)$, $(CN_\exists)$, and the carrier condition $(U3)$, the conformal skeleton and the conformal modular sector cannot be independently varied. More precisely:*

1. *the standard HSMI data determine a unique conformal skeleton;*
2. *the conformal skeleton, once normalized by the half-line identification above, determines the conformal modular sector;*
3. *therefore any admissible deformation that preserves one of these two structures must preserve the other.*

Proof. Item (1) is exactly Theorem 4.7.

For item (2), the Möbius net comes equipped with a positive-energy representation of $\mathrm{PSL}(2,\mathbb{R})$. Once the half-line normalization

$$\mathcal{A}_{\mathrm{GLW}}(0,+\infty) = \mathcal{A}_c, \qquad \mathcal{A}_{\mathrm{GLW}}(1,+\infty) = \mathcal{B}_{\mu_0}$$

is fixed, the corresponding dilation and translation subgroups recover the modular action and the positive translation generator associated with the inclusion. In other words, the conformal skeleton rigidly determines the conformal modular sector.

Item (3) is immediate. If an admissible deformation preserves the conformal modular sector, item (1) forces the conformal skeleton to remain fixed. Conversely, if it preserves the conformal skeleton, item (2) forces the modular sector to remain fixed. Hence the two structures are not independently deformable. □

This result is weak only in the sense that it stops at the conformal level. It does not yet recover metric scale. That is the purpose of Step I and Step II. But it already establishes the central conceptual point of Phase 1:

> causal/conformal geometry and modular time are not two separately specified ingredients;
> they are two readings of the same standard half-sided modular data.

4.11 What Phase 1 Has Actually Closed

It is useful to isolate the precise mathematical output of the section before moving on to metric scale.

1. HSMI is not separately assumed. It is a conditional theorem derived from the explicit interface (Amb) and dynamical axiom (C).
2. The second direction is no longer a global mystery. It is isolated into the existence-type conormality input $(CN_{\exists})$.
3. The conformal skeleton is no longer conjectural. It is reconstructed canonically by GLW from the fixed standard inclusion.
4. The first inseparability theorem is already in hand: conformal geometry and conformal modular data are mutually rigid.

What remains open after Phase 1 is not the existence of the conformal skeleton, but the recovery of metric scale from local modular data. That is why the next phase turns to the cross-ratio family and to the canonical local modular scale datum

$$\{K_b\}_{b\in(0,1)}.$$

5 From Conformal Skeleton to Full Geometry (Phase 2)

Phase 1 established that conformal geometry and conformal modular data are inseparable. But the conformal skeleton determines only the causal structure, the light cones. It does not determine distances. A conformal class of Lorentzian metrics contains infinitely many metrics related by Weyl rescaling. To reach the full inseparability statement, we must show that the metric scale is also locked to modular data.

This is the task of Phase 2. The strategy has three steps: identify the correct local geometric modulus, extract a canonical modular scale datum from it, and show that this datum determines the metric scale via a Type III-compatible quantum Fisher information metric.

5.1 Why Single-Interval Deformations Fail

The first instinct is to deform a single interval within the GLW net and track how the modular Hamiltonian changes. This fails for a precise reason.

On S^1, the group $\mathrm{PSL}(2,\mathbb{R})$ acts transitively on the space of intervals. Any single-interval deformation, such as moving one endpoint while fixing the other, is a Möbius transformation. Since the vacuum state ω is Möbius-invariant, pulling back the restricted state along such a deformation produces a constant family. The quantum Fisher information metric of a constant family is zero.

The lesson is that single-interval endpoint motion is a gauge orbit, not a geometric deformation. To obtain a nontrivial modulus, one needs at least four boundary points.

5.2 The Cross-Ratio as the Minimal Non-Möbius Modulus

Consider a nested double-interval configuration within the GLW net:

$$\mathcal{M}_{\text{in}}(b) \subset \mathcal{M}_{\text{out}}, \qquad b \in (0,1).$$

In the standard half-line gauge, this means

$$\mathcal{M}_{\text{out}} = \mathcal{A}_{\text{GLW}}(0,\infty), \qquad \mathcal{M}_{\text{in}}(b) = \mathcal{A}_{\text{GLW}}(b,1).$$

The four boundary points are $0, b, 1, \infty$. The Möbius group has three real parameters, so after gauge-fixing, exactly one real modulus survives: the cross-ratio

$$\chi = b.$$

This is the minimal non-Möbius geometric degree of freedom available in the GLW net. Varying b genuinely changes the relative position of the inner and outer algebras in a way that no Möbius transformation can undo.

5.3 The Canonical Modular Scale Datum

For each $b \in (0,1)$, the inclusion $\mathcal{M}_{\text{in}}(b) \subset \mathcal{M}_{\text{out}}$ determines a modular operator $\Delta_{\mathcal{M}_{\text{in}}(b),\Omega}$ via the Tomita-Takesaki theorem applied to $\mathcal{M}_{\text{in}}(b)$ with respect to the vacuum vector Ω.

Definition 5.1 (Canonical modular scale datum)**.**

$$K_b := -\ln \Delta_{\mathcal{M}_{\text{in}}(b),\Omega}, \qquad b \in (0,1).$$

The family $\{K_b\}_{b\in(0,1)}$ is the canonical local modular scale datum. It is determined entirely by the GLW net and the vacuum state. No additional choices are involved.

Proposition 5.2 (Cross-ratio is the minimal non-Möbius modulus)**.** *A nested double-interval configuration in the GLW net has exactly one Möbius-invariant modulus, the cross-ratio $\chi = b$. In particular, the family $\{K_b\}$ captures the minimal local modular scale information that is not already contained in the conformal skeleton.*

This is the theorem-level output of Step I. It requires no additional axioms beyond Phase 1.

5.4 The Chart Axiom and the Scale Shadow

To convert the modular scale datum into a Riemannian metric on the cross-ratio parameter space, we need to differentiate the family $\{K_b\}$ and evaluate a quadratic form. This requires regularity that is not automatic from the abstract GLW construction.

Axiom (LImp^1). For each reference point $b_0 \in (0,1)$, there exists a neighborhood U_{b_0} and a unitary family

$$W_b \in \mathcal{U}(\mathcal{M}_{\text{out}}), \qquad b \in U_{b_0},$$

satisfying:

1. $W_{b_0} = \mathbf{1}$;
2. $W_b \mathcal{M}_* W_b^* = \mathcal{M}_{\text{in}}(b)$ for a fixed reference algebra $\mathcal{M}_* = \mathcal{M}_{\text{in}}(b_0)$;
3. $b \mapsto \omega(W_b A W_b^*)$ is C^1 for each $A \in \mathcal{M}_*$;
4. the tangent vector $\dot{\phi}_{b_0}$ at $b = b_0$ is represented by the modular Hamiltonian variation $\dot{K}_{b_0} = \frac{d}{db}|_{b=b_0} K_b$ via the standard Araki correspondence.

This axiom packages three regularity conditions into one: the existence of a local implementer (CR1), C^1 smoothness of the pulled-back state family (CR2), and tangent compatibility with the modular Hamiltonian variation (CR3).

Axiom (CR-nd) (Non-degeneracy). The Araki-Hessian quadratic form $\mathbf{Q}_{\omega,b}$ is strictly positive on the modular Hamiltonian variation:

$$\mathbf{Q}_{\omega,b}(\partial_b K_b, \partial_b K_b) > 0, \qquad \forall\, b \in (0,1).$$

In the worked examples, this follows from the positivity of the stress-tensor two-point function. It fails only if the cross-ratio deformation produces no state-level change, a physically degenerate situation.

Under these two conditions, the scale shadow is a genuine Riemannian metric on the cross-ratio parameter space.

Theorem 5.3 (Scale shadow, conditional on (LImp^1) + (CR-nd))**.** *Under the Phase 1 axioms and the additional conditions $(LImp^1)$ and $(CR\text{-}nd)$, the canonical modular scale datum induces a Riemannian scale shadow*

$$h_b = \mathbf{Q}_{\omega,b}(\partial_b K_b, \partial_b K_b)\, db^2,$$

where $\mathbf{Q}_{\omega,b}$ is the positive quadratic form induced by the Araki relative entropy Hessian [1] at the state $\omega|_{\mathcal{M}_{\text{in}}(b)}$ via the $(LImp^1)$ chart.

Remark 5.4. In the special case where the chart is of exponential type (CR-exp), this reduces to the familiar variance formula

$$h_b = \text{Var}_\omega(\partial_b K_b)\, db^2.$$

The general formula via $\mathbf{Q}_{\omega,b}$ does not require this specialization.

5.5 Geometric Datum: Conformal Skeleton + Slice Scale

The conformal skeleton from Phase 1 determines the causal structure, equivalently the conformal causal structure. The scale shadow $\{h_b\}$ determines the complementary scale information along the cross-ratio slice family. At the abstract level established in this paper, the pair $(\text{Conf}, \{h_b\})$ is the full geometric datum. In the worked examples of Section 6, the cross-ratio parameter b naturally parametrizes the spatial position of the inner interval endpoint within the outer interval. The scale shadow h_b therefore directly defines a Riemannian metric on the spatial slice, and together with the conformal causal structure determines a Lorentzian metric.

Corollary 5.5 (Full geometric datum, 1+1 dimensions)**.** *In the 1+1 dimensional setting of the GLW net, the conformal skeleton* $\text{Conf}(\mathcal{N}, \omega)$ *together with the scale family* $\{h_b\}_{b\in(0,1)}$ *defines the full geometric datum*

$$\text{Geom}_{\text{full}}(\mathcal{N}, \omega) := \big(\text{Conf}(\mathcal{N}, \omega),\ \{h_b\}_{b\in(0,1)}\big).$$

In the worked examples of Section 6, the cross-ratio parameter b naturally parametrizes the spatial position of the inner interval endpoint within the outer interval. The scale shadow h_b therefore directly defines a Riemannian metric on the spatial slice, and together with the conformal causal structure determines a Lorentzian metric.

Remark 5.6. The step from scale shadow to Lorentzian metric relies on the model-level identification of the cross-ratio parameter with spatial position. At the abstract level, Corollary 5.5 provides a full geometric datum; the Lorentzian interpretation is verified in the worked examples but not proved in full generality. The extension to higher dimensions is an open problem (Section 10.2).

5.6 Forward and Bidirectional Full Inseparability

With the full geometric datum in hand, we can state the main theorems.

Definition 5.7 (Extended modular data)**.**

$$\text{Mod}_{\text{ext}}(\mathcal{N}, \omega) := \big(\text{Mod}_{\text{conf}}(\mathcal{N}, \omega),\ \{K_b\}_{b\in(0,1)}\big).$$

Two extended modular data $\text{Mod}_{\text{ext}}^{(1)}$ and $\text{Mod}_{\text{ext}}^{(2)}$ are identified if their conformal modular sectors agree and their canonical scale families $\{K_b\}$ agree up to the trivial equivalences specified in (SFa). All equalities of Mod_{ext} below are understood in this quotient sense.

Theorem 5.8 (Forward Full Inseparability)**.** *Under the Phase 1 axioms and* $(LImp^1)$ + $(CR\text{-}nd)$*, if two admissible objects have the same extended modular data, then they have the same full geometric datum:*

$$\text{Mod}_{\text{ext}}^{(1)} = \text{Mod}_{\text{ext}}^{(2)} \quad \Longrightarrow \quad \text{Geom}_{\text{full}}^{(1)} = \text{Geom}_{\text{full}}^{(2)}.$$

Proof. By Phase 1 Weak Inseparability, the conformal modular data determine the conformal skeleton. By Theorem 5.3, the canonical modular scale family $\{K_b\}$ determines the scale shadow $\{h_b\}$. By Corollary 5.4, the conformal skeleton and scale shadow together determine the full geometric datum. □

The forward direction says: you cannot change the full geometric datum without changing the modular data. But can you change the modular data without changing the full geometric datum? This requires an additional faithfulness condition.

Axiom (SFa) (Scale faithfulness). The canonical scale map $\{K_b\} \mapsto \{h_b\}$ is faithful within the admissible class: if $h_b^{(1)} = h_b^{(2)}$ for all b, then $K_b^{(1)} = K_b^{(2)}$ for all b, up to trivial equivalences that do not affect the scale map.

Theorem 5.9 (Forward result and conditional converse)**.** *Under the Phase 1 axioms,* $(LImp^1)$*,* $(CR\text{-}nd)$*, and* (SFa)*, the derived implication is*

$$\mathrm{Mod}_{\mathrm{ext}}^{(1)} = \mathrm{Mod}_{\mathrm{ext}}^{(2)} \quad \Longrightarrow \quad \mathrm{Geom}_{\mathrm{full}}^{(1)} = \mathrm{Geom}_{\mathrm{full}}^{(2)}.$$

If the additional scale-faithfulness axiom (SFa) *is supplied, then the converse implication also holds, and hence*

$$\mathrm{Geom}_{\mathrm{full}}^{(1)} = \mathrm{Geom}_{\mathrm{full}}^{(2)} \quad \Longleftrightarrow \quad \mathrm{Mod}_{\mathrm{ext}}^{(1)} = \mathrm{Mod}_{\mathrm{ext}}^{(2)}.$$

Proof. The displayed forward implication is Theorem 5.6. For the conditional converse: by the definition of $\mathrm{Geom}_{\mathrm{full}}$, equality of full geometries determines both the conformal skeleton and the scale family $\{h_b\}$. The conformal skeleton determines the conformal modular data by Phase 1 Weak Inseparability. The scale family determines the canonical modular scale family $\{K_b\}$ by (SFa). Together, $\mathrm{Mod}_{\mathrm{ext}}^{(1)} = \mathrm{Mod}_{\mathrm{ext}}^{(2)}$. □

The converse is therefore not an independently derived theorem: its injective step is exactly the model-level axiom (SFa). Under the identifications defended in Section 2, the mathematical counterpart of physical structure (the full geometric datum $\mathrm{Geom}_{\mathrm{full}}$) and the mathematical counterpart of algebraic modular time (extended modular data) cannot be independently specified within this conditional package. Modular flow supplies an algebraic time parameter only. Identifying that parameter with phenomenal or first-person time is an additional interpretive move, not a conclusion of Theorems 4.8 or 5.9.

6 Abstract Sufficient Conditions for $(LImp^1)$

The Phase 2 theorems are conditional on the chart axiom $(LImp^1)$, the non-degeneracy condition $(CR\text{-}nd)$, and the faithfulness condition (SFa). Of these three, $(LImp^1)$ is the most structurally demanding: it requires a local unitary implementer family with C^1 regularity and tangent compatibility. This section identifies two abstract routes for discharging $(LImp^1)$: a full discharge for Diff-covariant nets, and a decomposition of the remaining non-Diff burden into transportability, inner implementability, and regularity.

The conditions $(CR\text{-}nd)$ and (SFa) are not addressed here. They appear to depend on dynamical and spectral properties of the specific model (e.g., the positivity of the stress-tensor two-point function for $(CR\text{-}nd)$, and the injectivity of the variance map for (SFa)), and are better treated as model-level verifications rather than consequences of structural regularity.

6.1 Decomposition of $(LImp^1)$

It is useful to isolate the three independent requirements packaged into $(LImp^1)$:

1. **Local transport.** For each b near a reference point b_0, there exists an isomorphism $\alpha_b \colon \mathcal{M}_* \to \mathcal{M}_{\mathrm{in}}(b)$ that extends to an automorphism of $\mathcal{M}_{\mathrm{out}}$.

2. **Unitary implementability.** The automorphism α_b is inner in $\mathcal{M}_{\text{out}}$: there exists $W_b \in \mathcal{U}(\mathcal{M}_{\text{out}})$ with $\alpha_b = \text{Ad}(W_b)$.

3. C^1 **regularity and tangent compatibility.** The map $b \mapsto \omega(W_b A W_b^*)$ is C^1 for each $A \in \mathcal{M}_*$, and the tangent at b_0 is represented by the modular Hamiltonian variation $\dot{K}_{b_0}$ via the Araki correspondence.

Different sufficient conditions address different subsets of these requirements.

6.2 Why Split Property Alone Is Not Sufficient

The split property for the inclusion $\mathcal{M}_{\text{in}}(b) \subset \mathcal{M}_{\text{out}}$ guarantees the existence of a Type I factor $\mathcal{N}$ interpolating between the two algebras, and hence a local tensor product decomposition of the Hilbert space. This is relevant to requirement (2): in the presence of a split, the "annular region" between the inner and outer algebras has enough room to localize unitary implementers.

However, the split property does not by itself produce the transport of requirement (1). It tells you that the relative position of $\mathcal{M}_{\text{in}}(b)$ inside $\mathcal{M}_{\text{out}}$ is "standard" in a measure-theoretic sense, but it does not provide a canonical family of automorphisms mapping $\mathcal{M}_*$ to $\mathcal{M}_{\text{in}}(b)$ as b varies. That family must come from an external source: either a symmetry group acting on the net, or an explicit transportability assumption.

Nor does the split property address requirement (3). The C^1 regularity of the pulled-back state family and its tangent compatibility with the modular Hamiltonian variation are analytic conditions that depend on the smoothness of whatever generates the transport, not on the split structure.

In summary: split property assists with unitary implementability but does not generate transport or regularity. It is a facilitating condition, not a sufficient one.

6.3 Diff-Covariant Nets

The cleanest sufficient condition for $(LImp^1)$ is diffeomorphism covariance of the net.

Theorem 6.1 (Diff-covariance implies $(LImp^1)$)**.** *Let $I \mapsto \mathcal{A}(I)$ be a local net on S^1 (or on the half-line) that is covariant under a strongly continuous positive-energy representation of* Diff(S^1) *(respectively, of orientation-preserving diffeomorphisms of the half-line fixing the boundary). Then $(LImp^1)$ holds for the cross-ratio family $\{\mathcal{M}_{\text{in}}(b)\}_{b\in(0,1)}$ inside $\mathcal{M}_{\text{out}}$.*

Proof. Fix a reference point $b_0 \in (0,1)$ and set $\mathcal{M}_* = \mathcal{M}_{\text{in}}(b_0)$. For each b in a neighborhood of b_0, choose a smooth family of diffeomorphisms f_b satisfying $f_{b_0} = \text{id}$, $f_b(b_0) = b$, $f_b(1) = 1$, with support contained in the outer interval.

Transport. The diffeomorphism covariance of the net gives $f_b(\mathcal{M}_*) = \mathcal{M}_{\text{in}}(b)$, and since $\text{supp}(f_b)$ is contained in the outer interval, the corresponding automorphism α_b restricts to an automorphism of $\mathcal{M}_{\text{out}}$. This establishes requirement (1).

Unitary implementability. The positive-energy representation provides unitaries $U(f_b)$ implementing α_b. Since f_b is supported in the outer interval, $U(f_b) \in \mathcal{M}_{\text{out}}$ by Haag duality (or by the localization properties of the Diff representation). Set $W_b := U(f_b)$. This establishes requirement (2).

C^1 regularity. The map $b \mapsto f_b$ is smooth by construction. The strong continuity of the Diff representation ensures that $b \mapsto U(f_b)$ is strongly continuous, and the smoothness of f_b upgrades this to C^1 regularity of the pulled-back state family $b \mapsto \omega(W_b A W_b^*)$.

Tangent compatibility. The infinitesimal generator of the family $U(f_b)$ at $b = b_0$ is a smeared stress-energy tensor supported in the outer interval. By the relation between stress-energy deformations and modular Hamiltonian variations in conformal field theory [3], this generator represents the modular Hamiltonian variation $\dot{K}_{b_0}$ via the Araki correspondence. This establishes requirement (3).

Nontriviality. The diffeomorphisms f_b are not Möbius transformations (they fix boundary points that a Möbius transformation would move). Therefore the pulled-back state family is nonconstant, and the QFIM is nonzero. □

This theorem upgrades the Virasoro worked example of Section 7.2 from a model verification to an instance of a general result: any Diff-covariant conformal net automatically satisfies $(LImp^1)$.

6.4 Transportability-Based Sufficiency

For nets that are not Diff-covariant, a second sufficient condition can be formulated by combining an explicit transportability hypothesis with the injectivity of the outer algebra.

Definition 6.2 (Local transportability)**.** The cross-ratio family $\{\mathcal{M}_{\text{in}}(b)\}$ is *locally transportable* inside $\mathcal{M}_{\text{out}}$ if, for each $b_0 \in (0,1)$, there exists a neighborhood U_{b_0} and a family of automorphisms $\alpha_b \in \text{Aut}(\mathcal{M}_{\text{out}})$, $b \in U_{b_0}$, with $\alpha_{b_0} = \text{id}$ and $\alpha_b(\mathcal{M}_*) = \mathcal{M}_{\text{in}}(b)$.

Theorem 6.3 (Transportability-based sufficiency for $(LImp^1)$)**.** *Suppose:*

1. *the cross-ratio family is locally transportable inside $\mathcal{M}_{\text{out}}$;*
2. *$\mathcal{M}_{\text{out}}$ is an injective (equivalently, hyperfinite) Type III$_1$ factor;*
3. *the transport family $b \mapsto \alpha_b$ is C^1 in the sense that $b \mapsto \omega(\alpha_b(A))$ is C^1 for each $A \in \mathcal{M}_*$, with tangent compatible with the modular Hamiltonian variation.*

Then $(LImp^1)$ holds.

Proof. Requirement (1) of $(LImp^1)$ is exactly hypothesis (1). For requirement (2): by hypothesis (2) and the Connes theorem [14] ($\text{Out}(\mathcal{M}) = \{e\}$ for injective Type III$_1$ factors), every automorphism of $\mathcal{M}_{\text{out}}$ is inner. Hence $\alpha_b = \text{Ad}(W_b)$ for some $W_b \in \mathcal{U}(\mathcal{M}_{\text{out}})$. Requirement (3) is hypothesis (3). □

Remark 6.4. If the inclusion $\mathcal{M}_{\text{in}}(b) \subset \mathcal{M}_{\text{out}}$ additionally satisfies the split property, then the Type I interpolation allows W_b to be chosen with support localized in the annular region between the two algebras. This localization can improve the analytic control needed for the C^1 regularity of hypothesis (3), but it is not required for the theorem as stated.

It is important to be clear about what this theorem does and does not achieve. Hypotheses (1) and (3) —transportability and C^1 tangent compatibility —are already the two main substantive components of $(LImp^1)$ itself. The theorem's contribution is to identify the precise algebraic condition (injectivity of the outer factor) that bridges the gap between "transport exists as an automorphism" and "transport is unitarily implemented inside the outer algebra." For nets whose local algebras are known to be injective Type III$_1$ factors —which includes all local algebras in standard AQFT models —this bridge is automatic.

6.5 Status of the Three Phase 2 Conditions

The results of this section change the status of $(LImp^1)$ as follows:

- For Diff-covariant nets, $(LImp^1)$ is fully discharged by Theorem 6.1. No additional assumptions are needed.
- For non-Diff-covariant nets with injective Type III_1 local algebras, $(LImp^1)$ reduces to local transportability plus C^1 regularity (Theorem 6.3). The remaining burden is to verify that the cross-ratio family admits a smooth transport, which is a geometric property of the net rather than an abstract axiom.
- $(CR\text{-}nd)$ and (SFa) remain model-level verifications. They depend on dynamical and spectral properties (positivity of the stress-tensor two-point function, injectivity of the scale map) that are not captured by structural regularity conditions.

The net effect is that, for Diff-covariant nets, the Phase 2 conditionality is reduced from three independent axioms to two model-level verifications. For non-Diff-covariant nets, the situation is more modest: $(LImp^1)$ is decomposed and partially discharged, but the transportability and regularity inputs remain substantive.

7 Verification: Two Worked Examples

The abstract Phase 2 result is deliberately conditional. Of the three conditions, $(LImp^1)$ now has abstract sufficient conditions (Section 6): it is fully discharged for Diff-covariant nets, and for other nets the remaining burden is isolated to transportability and regularity. The remaining conditions $(CR\text{-}nd)$ and (SFa) are verified in concrete models. This section provides that verification in two structurally distinct examples.

The examples are chosen to do different kinds of work. The Rindler/free-field chiral half-line model shows that the Phase 2 package is realized in the same physical family that motivated much of the original HAFF intuition. The Virasoro-net example then shows that the construction is not a free-field accident and, in particular, that the chart axiom becomes especially natural once $\mathrm{Diff}(S^1)$ covariance is built in.

7.1 Worked Example I: Rindler / Free-Field Chiral Half-Line

The first example must be formulated carefully. The correct Phase 2 object is not the family of translated Rindler wedges. That family is useful as a sanity check for the Phase 1 half-sided modular structure, but it is the wrong object for the cross-ratio construction because global translations leave the vacuum invariant and thus produce a constant pulled-back state family. The correct model is the chiral half-line realization associated with the Rindler horizon.

We therefore take:

$$\mathcal{M}_{\text{out}} = \mathcal{A}(0,\infty), \qquad \mathcal{M}_{\text{in}}(b) = \mathcal{A}(b,1), \qquad b \in (0,1),$$

in the vacuum representation of a chiral free-field net. The canonical modular scale datum is then

$$K_b := -\ln \Delta_{\mathcal{M}_{\text{in}}(b),\Omega}.$$

7.1.1 Verification of $(LImp^1)$

In this model, the required local chart is supplied by the local diffeomorphism covariance of the free-field chiral net. Specifically, we assume covariance under orientation-preserving diffeomorphisms of the half-line that fix the boundary, implemented by unitaries in the corresponding local algebras. Fix a reference point $b_0 \in (0,1)$ and set

$$\mathcal{M}_* := \mathcal{A}(b_0, 1).$$

Choose a smooth local family of diffeomorphisms

$$f_b : (0,\infty) \to (0,\infty), \qquad b \in U_{b_0},$$

such that

1. $f_{b_0} = \mathrm{id}$;
2. $f_b(1) = 1$;
3. $f_b(b_0) = b$;
4. the support of f_b is contained in the outer half-line.

This covariance gives unitaries

$$W_b := U(f_b) \in \mathcal{U}(\mathcal{M}_{\text{out}})$$

with

$$W_b \mathcal{M}_* W_b^* = \mathcal{M}_{\text{in}}(b).$$

Because the deformation is genuinely boundary-fixed rather than Möbius, the family is nontrivial; because $b \mapsto f_b$ is smooth, the pulled-back state family is C^1; and because the infinitesimal generator is represented by the stress tensor, the tangent of the state family is compatible with the modular Hamiltonian variation. Thus $(LImp^1)$ holds in the model.

7.1.2 Verification of $(CR\text{-}nd)$

In a vacuum chiral conformal theory, the modular Hamiltonian of an interval (a,b) has the local CHM form [3]

$$K_{(a,b)} = 2\pi \int_a^b \frac{(x-a)(b-x)}{b-a}\, T_{00}(x)\, dx.$$

Specializing to the present family gives

$$K_b = 2\pi \int_b^1 \frac{(x-b)(1-x)}{1-b}\, T_{00}(x)\, dx,$$

and differentiation yields

$$\partial_b K_b = -2\pi \int_b^1 \frac{(1-x)^2}{(1-b)^2}\, T_{00}(x)\, dx.$$

The corresponding Araki-Hessian quadratic form is positive and nonzero, so the scale shadow is non-degenerate. More explicitly, one obtains

$$h_b = C_R (1-b)^{-2} db^2, \qquad C_R > 0.$$

Hence $(CR\text{-}nd)$ is satisfied.

7.1.3 Verification of (SFa)

In this model, faithfulness reduces to a one-dimensional injectivity statement. Since

$$h_b = C_R(1-b)^{-2}db^2$$

is strictly monotone in b, the metric profile determines the parameter b, and hence determines the corresponding modular datum K_b up to the trivial equivalences already allowed in the definition of (SFa). Therefore the scale map is faithful in the required sense.

The upshot is that the entire Phase 2 package is realized in the Rindler/free-field chiral half-line model:

$$(\text{LImp}^1), \quad (\text{CR-nd}), \quad (\text{SFa}).$$

7.2 Worked Example II: The Virasoro Net

The second example shows that the preceding mechanism is not tied to free fields. Let

$$I \longmapsto \mathcal{A}_{\text{Vir},c}(I)$$

be the Virasoro net at central charge c, with vacuum vector Ω. As in the first example, we work in the standard half-line gauge:

$$\mathcal{M}_{\text{out}} = \mathcal{A}_{\text{Vir},c}(0,\infty), \qquad \mathcal{M}_{\text{in}}(b) = \mathcal{A}_{\text{Vir},c}(b,1).$$

7.2.1 Verification of $(LImp^1)$

What was model-dependent in the first example becomes built in here. The Virasoro net is Diff(S^1)-covariant, so $(LImp^1)$ follows directly from Theorem 6.1. Concretely, one can choose a smooth boundary-fixed family

$$f_b \in \text{Diff}(S^1), \qquad b \in U_{b_0},$$

with

$$f_{b_0} = \text{id}, \qquad f_b(b_0) = b, \qquad f_b(1) = 1,$$

and support contained in the outer half-line. The positive-energy Diff(S^1) representation yields unitaries

$$W_b := U(f_b) \in \mathcal{U}(\mathcal{M}_{\text{out}})$$

such that

$$W_b \mathcal{A}(b_0,1) W_b^* = \mathcal{A}(b,1).$$

The nontriviality, smoothness, and tangent compatibility required by $(LImp^1)$ follow immediately from the non-Möbius character of f_b, the smoothness of the Diff(S^1) action, and the fact that the infinitesimal action is generated by the Virasoro stress tensor. Thus $(LImp^1)$ holds here in a cleaner and more canonical way than in the free-field model.

7.2.2 Verification of $(CR\text{-}nd)$

The interval modular Hamiltonian again has the CHM form

$$K_b = 2\pi \int_b^1 \frac{(x-b)(1-x)}{1-b}\, T(x)\, dx,$$

where T is now the Virasoro stress tensor. Differentiating gives the same structural formula as before:

$$\partial_b K_b = -2\pi \int_b^1 \frac{(1-x)^2}{(1-b)^2}\, T(x)\, dx.$$

Using the Virasoro two-point function

$$\langle T(x)T(y)\rangle \sim \frac{c/2}{(x-y)^4},$$

one obtains, after standard regularization, a positive finite coefficient

$$\mathbf{Q}_{\omega,b}(\partial_b K_b, \partial_b K_b) = \frac{C_{\text{Vir}}(c)}{(1-b)^2}, \qquad C_{\text{Vir}}(c) > 0.$$

Hence

$$h_b = C_{\text{Vir}}(c)(1-b)^{-2} db^2,$$

and $(CR\text{-}nd)$ holds.

7.2.3 Verification of (SFa)

Exactly as in the first example, the scale profile is strictly injective in the cross-ratio parameter. Therefore the full scale family determines b, and hence determines the local modular datum up to the same trivial equivalences. Thus (SFa) is satisfied in the Virasoro model as well.

7.3 Comparison and Universal Scaling

The significance of the two examples lies less in their individual computations than in what they share.

1. Both models realize the same abstract Phase 2 package:
$$(\text{LImp}^1) + (\text{CR-nd}) + (\text{SFa}).$$
2. Both produce the same scale profile:
$$h_b = C(1-b)^{-2} db^2.$$
3. The model dependence enters only through the coefficient:
   - in the free-field model through the normalization of the stress-tensor sector;
   - in the Virasoro model through the central charge c.

This is the strongest available evidence, within the present paper, that the scale shadow is not a model-specific artifact. What appears to be universal is the cross-ratio scaling law itself, which is fixed by the CHM structure of interval modular Hamiltonians. What remains model-sensitive is only the overall constant.

From the point of view of the paper's philosophical thesis, this matters because it shows that the full inseparability mechanism is not tied to one special solvable example. The rigidity claim is abstract; the examples show that the additional Phase 2 inputs are concretely realizable in more than one structural setting, and that the theorem's conditions arise naturally where one would most expect them to arise.

8 Implications for the Hard Problem

For the purposes of this section, we take the mathematical results of the previous sections as given. At the conformal level, Theorem 4.8 already shows that conformal geometry and conformal modular data are not independently deformable. The later Phase 2 results strengthen this to the full scale-sensitive level by tying the metric scale to the canonical modular scale datum. The philosophical question is therefore no longer whether one can attach a speculative interpretation to a formal analogy. It is whether the standard formulation of the hard problem survives once its key presupposition is written in mathematically precise terms.

8.1 The Failure of the Presupposition

Section 1 identified the hidden presupposition of the hard problem: independent describability. Section 2 translated it into independent deformability. The inseparability theorems of Sections 4 and 5 now supply the answer.

At the Phase 1 level, Theorem 4.8 shows that conformal geometry and conformal modular data are already locked together. At the full level, Theorem 5.7 shows that, under $(LImp^1)$, $(CR\text{-}nd)$, and (SFa), the full geometric datum and the extended modular data cannot be independently varied. Conditional on the identifications defended in Section 2, the presupposition is not available at either level.

The strongest claim justified by the theorem is therefore not that the hard problem has been "solved" in the ordinary sense. It is that its canonical production form is ill-posed. The question "how does matter produce mind?" asks for a derivation across a divide that, in the present framework, is not mathematically available as an independent divide. One cannot first specify the geometric-physical side and only afterwards ask how the first-person side appears. They are two structurally inseparable readings of the same algebraic datum $(\mathcal{A}, \omega)$.

8.2 Relocating the Explanatory Gap

Once the production picture is rejected, a residual question remains. Even if geometry and modular flow are inseparable, why should a given modular invariant correspond to a given phenomenal quality? This is legitimate, but it is not the original hard problem. It concerns the interpretive reach of an observer embedded in the very structure being described.

The proper location of the remaining gap is not between physical and experiential structure as two externally juxtaposed domains. A possible research direction places it within the self-description limits of a finite agent modeled by $(\mathcal{A}, \omega)$. Paper G's discussion of self-reference [9] motivates this direction, but it does not prove a generic fixed-point no-go or an exhaustive regress-versus-collapse dichotomy. No theorem of the present paper therefore infers a phenomenal limitation from Paper G.

On that interpretation, the explanatory gap may be treated as structural rather than merely technical [7]. Establishing such a horizon would require a separate formal bridge from algebraic self-reference to phenomenal description; the analogy alone does not supply it.

This changes what counts as philosophical success. The goal is no longer to show how an independently specified physical world outputs experience. It is to understand why an embedded agent necessarily encounters a limit in its attempt to completely internalize the relation between structural invariants and lived articulation. The residual gap is relocated and reclassified, not removed.

8.3 Comparison with Existing Positions

The closest philosophical relatives of the present framework are positions that deny a brute separation between mind and world. Even so, the relation is exact only up to a point. The following comparison is necessarily compressed; each position listed has significant internal diversity, and the characterizations below refer to the canonical or most common formulation.

Position	Relation to the present paper
Physicalism / functionalism	Shares the desire for structural rigor, but typically preserves an explanatory asymmetry: physical facts are taken as independently specifiable, and consciousness must be shown to follow. The present theorem denies that asymmetry in its standard form.
Panpsychism	Avoids emergence by treating experience as basic, but inherits the combination problem. The present framework does not posit basic micro-experiences; it treats modular structure and geometry as co-emergent readings of one algebraic organization.
Dual-aspect monism	The nearest metaphysical relative. The decisive difference is that here the dual-aspect claim is not left at the level of metaphysical slogan; it is tied to a concrete inseparability theorem.
IIT	Also seeks a mathematically articulated bridge, but introduces an additional quantity Φ as a privileged measure. The present route instead treats inseparability as internal to modular and geometric structure, not as the effect of an extra consciousness functional.
Illusionism	Rejects the hard problem by denying that phenomenal consciousness needs the explanatory role usually assigned to it. The present framework does not deny experience. It denies the independent specification presupposition that makes experience look like a separately generated add-on.
Russellian monism	Very close in spirit insofar as it challenges the exhaustiveness of third-person structure. But Russellian monism often speaks of an intrinsic categorical basis behind structural physics. The present framework is more austere: both geometry and first-person time are readings of one algebraic structure, without appeal to a hidden intrinsic substrate beyond that structure.

The upshot is that the present work should not be advertised as a simple algebraic version of any single existing position. It is closest to a theorem-backed dual-aspect view, but even that label is only approximate. The real novelty lies in replacing a metaphysical balance claim with a rigidity statement.

A structural correspondence with the Yogacara fourfold analysis is recorded in Appendix A.

9 Broader Significance

The philosophical argument of the preceding sections is the primary contribution of this paper. The mathematical and physical content, however, has independent interest that is worth recording briefly.

9.1 Mathematical Significance

The Wiesbrock-GLW programme [17, 5] established that half-sided modular inclusions reconstruct conformal symmetry. What remained open was the passage from conformal skeleton to metric scale using only modular data. The present paper isolates a precise conditional route to a full geometric datum in 1+1 dimensions: the cross-ratio modular scale datum $\{K_b\}$ serves as the canonical non-gauge geometric modulus, and the Araki relative entropy Hessian [1] serves as the Type III-compatible QFIM that converts it into a Riemannian scale shadow. The route is conditional on $(LImp^1)$, $(CR\text{-}nd)$, and (SFa), all three of which are verified in two structurally distinct models. For Diff-covariant nets, Section 6 further shows that $(LImp^1)$ is automatic, leaving only $(CR\text{-}nd)$ and (SFa) as model-level verifications. In the worked examples, where the cross-ratio parameter naturally parametrizes the spatial slice, this datum assembles into the Lorentzian metric.

The universal scaling law $h_b \sim (1-b)^{-2}db^2$, shared by both worked examples, is a structural consequence of the Casini-Huerta-Myers form [3] of interval modular Hamiltonians. It is not a free-field accident. The model-dependent information enters only through the overall constant, which encodes data such as the central charge.

9.2 Physical Significance

The slogan "it from qubit" and the ER=EPR programme assert that entanglement structure determines spacetime geometry [11, 13, 16, 12]. These claims have so far been tied to the AdS/CFT correspondence. The present result establishes a version of the same thesis without assuming holographic duality. The axiom package selected by $(U3)$, together with the axiom package $(U1)+(C)+(Amb)+(CN_{\exists})+(LImp^1)+(CR\text{-}nd)+(SFa)$, makes no reference to anti-de Sitter space, conformal field theory on a boundary, or bulk-boundary correspondence. Inseparability is a property of Type III_1 algebraic structure under appropriate regularity [18], not a special consequence of holography.

This also resolves two open issues from the earlier HAFF programme (Paper D) [8]. The non-uniqueness of the map from algebraic deformations to state deformations is absorbed by the cross-ratio datum and the chart axiom, which together fix the relevant deformation direction. The signature mismatch between the positive-definite QFIM and the Lorentzian metric is resolved by restricting the QFIM to spatial slices and letting the conformal skeleton supply the causal structure.

10 Explicit Non-Claims and Open Problems

10.1 What This Paper Does Not Claim

1. It does not claim to have solved the hard problem of consciousness. It claims to have shown that the standard production formulation rests on a presupposition that is not available in the relevant mathematical framework.

2. It does not claim that modular flow "is" consciousness. The identification in Section 2 is a structural correspondence: modular flow plays the mathematical role that first-person temporal structure plays in the philosophical debate.

3. It does not claim that the inseparability theorem holds unconditionally for all Type III_1 factors. Phase 1 is unconditional within the stated axiom class. Phase 2 is conditional

on $(LImp^1)$, $(CR\text{-}nd)$, and (SFa). For Diff-covariant nets, $(LImp^1)$ is discharged by Theorem 6.1; the remaining two conditions are verified in two models but not proved in full generality.

4. It does not claim that the Buddhist correspondence in Section 7.4 participates in any proof. It is included as a structural comparison, not as evidence.

5. It does not claim that the results extend to dimensions higher than 1+1. The GLW construction produces a Möbius net on S^1. Higher-dimensional generalizations require additional HSMI directions and remain open.

6. It does not claim that consciousness is a quantum effect. The argument concerns the structural relationship between geometry and modular flow at the level of algebraic quantum theory, not a physical mechanism involving quantum coherence in biological systems.

10.2 Open Problems

1. Abstract sufficient conditions for $(CR\text{-}nd)$ and (SFa). Section 6 provides sufficient conditions for $(LImp^1)$ (Diff-covariance or split-assisted transportability), but the remaining two conditions are still verified only in specific models. A general criterion for $(CR\text{-}nd)$, perhaps in terms of spectral properties of the stress-energy tensor, would further reduce the conditionality.

2. Foliation compatibility beyond 1+1 dimensions. In the present 1+1 dimensional setting, the cross-ratio family is a single-parameter family and the assembly is straightforward. In higher dimensions, the scale data from different cross-ratio slices must satisfy nontrivial mutual consistency conditions. Formulating and verifying these conditions is an open problem tied to the higher-dimensional extension.

3. Higher-dimensional extension. In $d > 1 + 1$, the conformal group is larger and the cross-ratio construction must be replaced by a higher-dimensional analogue. Whether the inseparability mechanism survives this extension is the most important structural open question.

4. A formal self-description bridge. Paper G currently supplies philosophical motivation, not a fixed-point no-go theorem. A separate model with a typed syntax/observable interface and an explicit capacity functional would be required before modular invariants could yield a quantitative self-description bound.

5. The relationship between the present framework and Integrated Information Theory. Both seek mathematically articulated accounts of consciousness. Whether the inseparability theorem implies constraints on Φ or vice versa is unexplored.

A Yogacara Fourfold Correspondence

The Yogacara fourfold analysis provides a structural correspondence for the architecture that emerges in this paper. The correspondence does not enter any proof; it is included as a disciplined interpretive comparison that becomes newly intelligible once the mathematical theorem is in place. Each position listed below has significant internal diversity within the Buddhist philosophical tradition; the characterizations refer to the canonical formulation.

Layer	Phenomenological role	Mathematical counterpart
相分	the presented object-side	emergent geometry Geom
见分	the act-side of cognition	modular flow σ_t^{ω}
自证分	reflexive awareness	the self-calibrating informational loop of T-DOME Paper III [10]
证自证分	the boundary of reflexivity	the incompleteness limit of Paper G [9]

The structural fit is that object-side and act-side are not heterogeneous substances but internally linked aspects of a single cognitive event; reflexivity does not continue indefinitely into an infinite hierarchy, but closes through a higher-order boundary condition. This is exactly the pattern suggested by the inseparability framework: geometry and modular flow are inseparable at the base level, and reflexive closure is marked by incompleteness rather than by defect.

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